Computer Networks
Fall 2024/25
SOLUTION
Exercise-5-solution

Problem 1

Step	Source IP	Destination IP	Source Port	Destination Port
1 – from host	10.1.1.1	200.82.83.84	3333	8080
2 – from NATted router	41.42.43.44	200.82.83.84	**6688**	8080
3 – from server	200.82.83.84	41.42.43.44	8080	6688
4 – from NATted router	200.82.83.84	10.1.1.1	8080	3333

Internet side	LAN side
41.42.43.44, 6688	10.1.1.1, 3333

Problem 2

Answer:

1. The number of edges that become core edges is **7,** and the core edges are **1,3,4.6,7,8,9.**

2. The level is **3.**

Problem 3 & 4

Answer to question 3.

1. Hubs do not have tables.

Box1	Box4
A,1	A,3
B,2	C,3
C,3	D,3
D,3	E,3
E,3	F,3
F,3	G,1
G,3	H,2

2. **16** (4 ARP queries + 4 ARP responses + 8 IP packets)
3. **4** (C to D; D to C; E to F; F to E)
4. **4** (A to B; C to D; E to F; G to H)
5. **6** (A to B; C to D; D to C; E to F; F to E; G to H)
6. **4** (4 ARP-Broadcast from A,C,E,G)
7. **0**

Answer to question 4.

1. Hubs do not have tables.

Box1	Box4
A,1	A,3
B,2	B,3
C,3	C,3
D,3	D,3
E,3	E,3
F,3	F,1
G,3	G,1
H,3	H,2

2. **24** (8 ARP queries + 8 ARP responses + 8 IP packets)
3. **4** (C to D; D to C; E to F; F to E)
4. **4** (A to B; C to D; E to F; G to H)
5. **6** (A to B; C to D; D to C; E to F; F to E; G to H)
6. **8** (8 ARP-Broadcast from A,B,C,D,E,F,G,H)
7. **0**